2 Theoretical Setup

2.1 Data Distributions

Consider an iid sample from $\mathcal{D}_1 = \{(x_i, y_i) \mid 1 \le i \le n_1\}$ of size n_1 from the true data distribution P_1 and an independent iid sample $\mathcal{D}_2 = \{(x_i, y_i) \mid n_1 + 1 \le i \le n\}$ of size n_2 from another data distribution P_2 (which we shall hereafter call the *synthetic data distribution*), where $n := n_1 + n_2$ is the total amount of training data. Here, $P_k = P_{\Sigma_k, w_k^*, \sigma_k^2}$ is the distribution on $\mathbb{R}^d \times \mathbb{R}$ given by

$$\begin{aligned} &\textbf{(Features) } x \sim N(0, \Sigma_k), \\ &\textbf{(Labels) } y = x^\top w_k^* + \epsilon, \text{ with } \epsilon \sim N(0, \sigma_k^2) \text{ independent of } x. \end{aligned} \tag{1}$$

Each Σ_k is a $d \times d$ positive-definite covariance matrix which captures the intrinsic variations of the input feature vector x. The σ_k's control the level of label noise in each distribution.

Structure of the Label Shift. For conciseness, we will assume the following priors on the w_k^*'s

- *True labelling function:* $w_1^* \sim N(0, \Gamma)$,
- *Mismatch between real and synthetic:* $\delta := w_2^* - w_1^* \sim N(0, \Delta)$, independent of w_1^*,

for some $d \times d$ positive-semidefinite matrices Γ and Δ.

Remark 1. *To ease the presentation of our results, we shall assume that the matrices Σ_1, Σ_2, Γ, and Δ are diagonal matrices, and therefore commute. Furthermore, except otherwise explicitly stated, we shall assume equal covariance matrices, and take $\Sigma_1 = \Sigma_2 = \Sigma$ as in Dohmatob et al. (2024a).*

The matrix Γ captures the structure of the ground-truth labelling function in the real / test distribution P_1. Together with the label-noise levels σ_1^2 and σ_2^2, the matrix $\Delta = \operatorname{cov}(w_2^* - w_1^*)$ captures the covariance structure of the disparity between the true data distribution P_1 and the synthetic data distribution P_2 regarding the conditional distribution $p(y|x)$; the marginal distribution of x stays the same under P_1 and P_2 due the assumption $\Sigma_1 = \Sigma_2 = \Sigma$. For example, the self-consuming-loops setup of Dohmatob et al. (2024a) corresponds to taking Δ proportional to the precision matrix of the input features Σ^{-1}. Thus, the size of the fluctuations of each component δ_j of the difference $w_2^* - w_1^*$ is inversely proportional to the standard deviation of the corresponding feature. Another important setup is the case where the fluctuations are isotropic, i.e taking $\Delta \propto I_d$.

Quality of Synthetic Data. Due to the a priori general structure of Δ, the label corresponding to an input x will be different for both distributions, even in the absence of label-noise. On average, the L_2-norm of this difference is $\mathbb{E}_{w_1^*, w_2^*} \mathbb{E}_{x \sim N(0,\Sigma)} [(x^\top w_1^* - x^\top w_2^*)^2] = \operatorname{tr} \Sigma\Delta$. We therefore define

Definition 1. *The quality of synthetic data is defined as $c^2(\Delta) = (1/d) \operatorname{tr} \Sigma\Delta$, which captures the disparity between the synthetic data distribution P_2 and the real data distribution P_1 (small values of $c^2(\Delta)$ are better). For example, if $\Delta = c^2\Sigma^{-1}$ as in Dohmatob et al. (2024a), then $c^2(\Delta) = c^2$.*

2.2 Models and Performance Measure

Given this training data, the goal of a learner is to construct an estimator $\widehat{w}$. This can be seen as a linear model from $x \mapsto x^\top \widehat{w}$. Evaluated on the real / true data distribution P_1 (which coincides with the distribution from which the real component $\mathcal{D}_1$ of the training dataset $\mathcal{D}$ is drawn), the test error of a model $\widehat{f} : \mathbb{R}^d \to \mathbb{R}$ is defined by

$$E_{test}(\widehat{f}) = \mathbb{E}_{\mathcal{D}} \mathbb{E}_{x \sim N(0,\Sigma_1)} [(\widehat{f}(x) - x^\top w_1^*)^2]. \tag{2}$$

This will be our main object of study, for different models $\widehat{f}$. The outermost expectation $\mathbb{E}_{\mathcal{D}}$ is to quench the randomness in the training dataset $\mathcal{D}$ used to train the model.

We consider two families of analytically tractable models: (1) classical linear models obtained via penalized regression in the input space, and (2) models obtained via penalized regression in a feature space given by random projections. The latter allows us to study the role of model size in model collapse, by varying the output dimension of the random projection mapping. This output dimension m controls the size of a neural network in a simplified regime (Maloney et al., 2022; Bach, 2023).

(1) Classical Linear Model. We start with a setup motivated by Dohmatob et al. (2024a). We are interested in the penalized linear model (ridge) $\widehat{f}_{CL} : x \mapsto x^\top \widehat{w}$ with parameter vector $\widehat{w}$ given by

$$\widehat{w} = \arg\min_{w \in \mathbb{R}^d} \frac{1}{n} \sum_{i=1}^{n} (x_i^\top w - y_i)^2 + \lambda \|w\|^2, \tag{3}$$

trained on the total dataset $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$. Of course, the unregularized limit $\lambda \to 0^+$ corresponds to ordinary least-squares (OLS). We shall work in the following so-called proportionate scaling limit

(Proportionate Scaling Limit for Classical Linear Model) For fixed $\phi \in (0, \infty), p_2 \in (0, 1)$,

$$d, n, n_1, n_2 \to \infty, \quad n_2/n \to p_2, \quad n_1/n \to p_1 = 1 - p_2, \quad d/n \to \phi. \tag{4}$$

The extreme cases $p_1 \to 0^+$ and $p_2 \to 0^+$ correspond to training on only synthetic (resp. real) data. In particular, $p_1 \to 0^+$ corresponds to the setting considered in Dohmatob et al. (2024a). Note that in the isotropic setting where $\Sigma \propto I_d$, ϕ controls the speed of learning on clean data. Indeed, for small ϕ, the scaling law in this case is known (Hastie et al., 2022) to be $E_{test} \simeq \sigma_1^2 \phi + O(\phi^2)$. As we shall see (Corollary 1), this scaling law gets deformed in the presence of synthetic data in the training dataset, leading to model collapse.

(2) Random Projections Model. We consider neural networks in a simplified regime which can be approximated via random projections (Maloney et al., 2022; Bach, 2023), i.e $f(x) = v^\top S x$. Here, S is a $d \times m$ random matrix with iid entries from $N(0, 1/d)$; it maps an input-vector $x \in \mathbb{R}^d$ to a random feature vector $z = \Phi(x) := Sx \in \mathbb{R}^m$. Only the "read-out" weights $v \in \mathbb{R}^k$ are learned, by fitting on the dataset $\mathcal{D}$. Consider the model $\widehat{f}_{RP} : x \mapsto \Phi(x)^\top \widehat{v}$, where $\widehat{v}$ is given by

$$\widehat{v} = \arg\min_{v \in \mathbb{R}^m} \frac{1}{n} \sum_{i=1}^{n} (\Phi(x_i)^\top v - y_i)^2 + \lambda \|v\|_2^2. \tag{5}$$

Note that such a simplified neural network model has been proposed in the literature as a theoretical testbed for studying intriguing properties of neural networks, like scaling laws (Maloney et al., 2022) and double-descent (Bach, 2023). Also see Section 1.2. It can be shown that the extreme case $m/n \to \infty$ reduces to the classical linear model.

We shall work in the following asymptotic regime:

(Proportionate Scaling Limit for Random Projections Model)

$$d, m, n, n_1, n_2 \to \infty, \quad n_1/n \to p_1,\ n_2/n \to p_2, \quad d/n \to \phi,\ m/d \to \gamma,\ m/n \to \psi, \tag{6}$$

for some constants $\phi, \gamma, \psi \in (0, \infty)$ and $p_1, p_2 \in (0, 1)$, with $p_1 + p_2 = 1$ and $\psi = \phi\gamma$.

Note that the ratio $\psi/\phi \simeq md$ captures the size of the network, though the number of trainable parameters (the read-out layer) is $m \simeq \gamma d$.

3 A New Bias-Variance Decomposition and the Emergence of Strong Model Collapse

3.1 Classical Linear Models

We begin with an analysis of the test error $E_{test}(\widehat{f}_{CL})$ for the classical linear model defined in (3) trained on a mixture of synthetic and true / real data, but evaluated on test data from the true data distribution only. We will establish a new bias-variance decomposition with an additional term which quantitatively reveals the emergence of model collapse (Shumailov et al., 2023, 2024).

Let us first recall some standard notations. Denote by $\kappa = \kappa(n, \lambda; \Sigma)$ the unique positive solution to the fixed-point equation

$$\kappa - \lambda = \kappa\, \mathrm{df}_1(\kappa; \Sigma)/n, \text{ with } \mathrm{df}_k(t; \Sigma) := \mathrm{tr}\, \Sigma^k (\Sigma + tI_d)^{-k}. \tag{7}$$

Also define $u = u(n, \lambda; \Sigma) \geq 0$ as follows

$$u := \frac{\mathrm{df}_2(\kappa; \Sigma)/n}{1 - \mathrm{df}_2(\kappa; \Sigma)/n}. \tag{8}$$

The following result (proved in the appendix, alongside all other theoretical results in this work) will be exploited in the sequel to show that the use of synthetic data in model training can lead to catastrophic effects regarding test error.

Theorem 1. *Define $\sigma^2 := p_1\sigma_1^2 + p_2\sigma_2^2$ and let $\kappa, u \geq 0$ be as previously constructed. In the proportionate scaling limit* (4)*, the test error w.r.t the true data distribution P_1, of the classical linear model $\widehat{f}_{CL}$ defined in* (3) *is given by $E_{test}(\widehat{f}_{CL}) \simeq \overline{E} + \zeta$, with*

$$\overline{E} = B + V, \quad V = \sigma^2 \frac{\mathrm{df}_2(\kappa; \Sigma)/n}{1 - \mathrm{df}_2(\kappa; \Sigma)/n}, \quad B = \kappa^2 \frac{\operatorname{tr} \Gamma\Sigma(\Sigma + \kappa I_d)^{-2}}{1 - \mathrm{df}_2(\kappa; \Sigma)/n}, \tag{9}$$

$$\zeta = p_2^2 \cdot (1 + p_1 u) \operatorname{tr} \Delta\Sigma^3(\Sigma + \kappa I_d)^{-2} + p_2 u \operatorname{tr} \Delta\Sigma(p_1\Sigma + \kappa I_d)^2(\Sigma + \kappa I_d)^{-2}. \tag{10}$$

Note that for $\Delta = 0$ (i.e $w_2^* = w_1^*$), which corresponds to assuming that the real data and the surrogate data have the same distribution, the above theorem gives $E_{test}(\widehat{f}_{CL}) \simeq \overline{E} \simeq B + V$ which is the classical bias-variance decomposition (Hastie et al., 2022; Richards et al., 2021) for ridge regression on n samples from the distribution $P_{\Sigma, w_1^*, \sigma^2}$. The extra term ζ appearing in Theorem 1 is responsible for model collapse! In Appendix B.2, we show how Theorem 1 recovers the main results of Dohmatob et al. (2024a) for special choices of the displacement matrix Δ.

Strong Model Collapse. In particular, in the "scaling laws" regime where $\phi \to 0^+$, it holds that $\zeta \simeq p_2^2 \operatorname{tr} \Delta$. In this case, if $\operatorname{tr} \Delta$ remains bounded away from zero, then so is ζ unless $p_2 \to 0^+$, i.e we discard all synthetic data from the training dataset. This is *strong* model collapse. It hints that model collapse as exposed by Shumailov et al. (2023, 2024); Hataya et al. (2023); Martínez et al. (2023a,b); Bohacek & Farid (2023); Briesch et al. (2023); Guo et al. (2023) cannot be fixed by naively mixing synthetic and real data during training. We show in Section 3.2 that this observation continues to hold in the setting of random projections model $\widehat{f}_{RP}$ defined in (5). Finally, in Section 5 we study what happens when the synthetic data and real data are strategically mixed during training.

Proving Theorem 1. It turns out that the analysis of the classical linear model's test error $E_{test}(\widehat{f}_{CL})$ in Theorem 1 amounts to the analysis of the trace of rational functions of sums of random matrices. Although the limiting spectral density of sums of random matrices is a classical computation using subordination techniques (Marčenko & Pastur, 1967; Kargin, 2015), this is not enough for the full analysis; a more involved analysis is required. This difficulty will be even greater in the setting of random projections $\widehat{f}_{RP}$. To the rescue, in Appendix D we shall employ operator-valued free probability theory (OVFPT) to compute the exact high-dimensional limits of such quantities. The tools of OVFPT have been used in the recent machine learning theory literature to obtain precise asymptotics for the test error of neural networks (trained on real data only) in various linearized settings (Adlam & Pennington, 2020; Tripuraneni et al., 2021; Lee et al., 2023).

Example: The Isotropic Case. To help unpack Theorem 1, consider the following concrete setup

$$\Sigma = I_d, \quad \Gamma = (r^2/d)I_d, \quad \Delta = (c^2/d)I_d,$$

for some constants $r, c > 0$. The constant c^2 captures how close the distribution of the synthetic data P_1 is to the distribution of the real data P_1; thus it captures the quality of the synthetic data. This gives $u \simeq \phi/((1+\kappa)^2 - \phi)$, where $\kappa > 0$ uniquely satisfies the fixed-point equation $\kappa - \lambda = \kappa\phi/(1+\kappa)$; this is a quadratic equation that characterizes the well-known Marchenko-Pastur law (Marčenko & Pastur, 1967). The quantities appearing in the formulae presented in Theorem 1 then take the following simple forms: $V = \sigma^2\phi/((1+\kappa)^2 - \phi)$ and $B = \kappa^2 r^2/((1+\kappa)^2 - \phi)$, and

$$\zeta = \left(p_2(1 + p_1 u) + (p_1 + \kappa)^2 u\right) p_2 c^2/(1+\kappa)^2.$$

In particular, in the unregularized limit $\lambda \to 0^+$ corresponding to OLS, we get $\kappa \to (\phi - 1)_+$. To further make things concrete, consider the under-parametrized case where $\phi \in (0, 1)$ in the proportionate scaling regime (4). The over-parametrized case $\phi \in (1, \infty)$ is treated in Appendix B.1. We deduce the following corollary to Theorem 1.